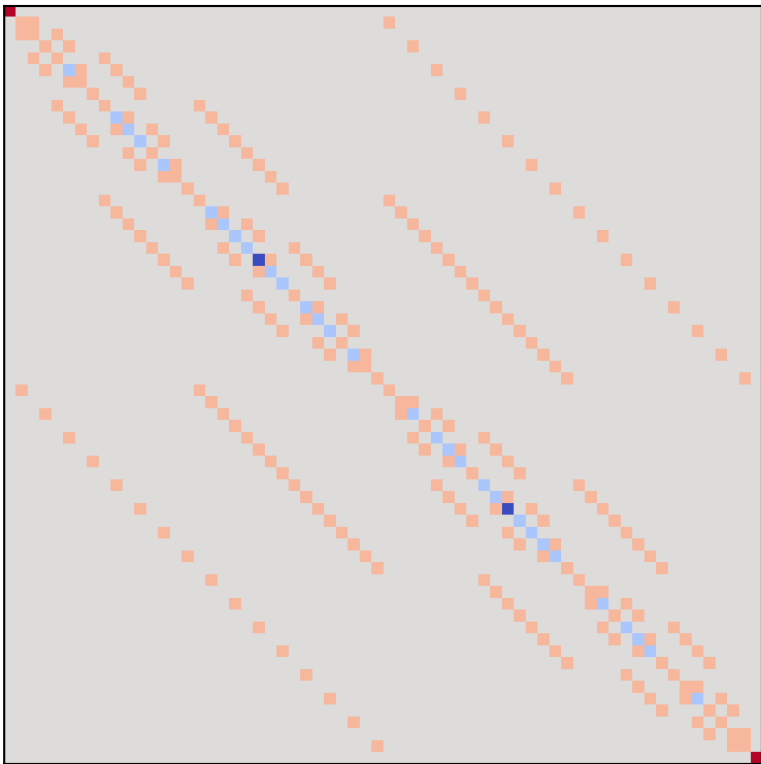


(i) Computational basis  $\{|s_1 s_2 \cdots s_N\rangle, s_i \in \{\uparrow, \downarrow\}\}$



(ii) Basis ordered by total magnetization  $S_{\text{tot}}^z$

